Introduction to Deep Learning

Session 9: Convolutional Neural Networks

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Lecture Overview

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1. Why Not MLPs for Images?
2. Convolution Operation
3. Pooling
4. CNN Architecture
5. Classic Architectures
6. Modern Details
7. Summary

Images as Data

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Image: tensor of shape $H \times W \times C$

- H, W — height and width in pixels
- C — channels (1 for grayscale, 3 for RGB)

Typical sizes:

Dataset	Size	Input dimensions
MNIST	$28 \times 28 \times 1$	784
CIFAR-10	$32 \times 32 \times 3$	3 072
ImageNet	$224 \times 224 \times 3$	150 528

MLP approach: flatten image to vector $\mathbf{x} \in \mathbb{R}^{H \cdot W \cdot C}$, apply fully connected layers

This leads to fundamental problems

Parameter Explosion

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MLP on ImageNet images: input dimension = $224 \times 224 \times 3 = 150\,528$

First hidden layer with 1024 units:

$$150\,528 \times 1024 \approx 154 \text{ million parameters} \quad \text{(one layer only)}$$

For comparison:

- ResNet-50 (entire network): ~25 million parameters
- VGG-16 (entire network): ~138 million parameters

Consequences:

- Enormous memory and compute requirements
- Massive overfitting — far more parameters than training examples
- Impractical to train even with modern hardware

No Spatial Structure

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MLP treats each pixel independently — no notion of spatial neighbourhood

Problem 1: no translation invariance

- Cat in top-left corner activates completely different weights than cat in bottom-right
- Same concept must be learned independently at every location
- $H \times W$ times more parameters needed

Problem 2: no local structure

- Nearby pixels are highly correlated — edges, textures, shapes span local regions
- MLP ignores this — pixel $(0, 0)$ and pixel $(223, 223)$ treated identically
- Spatial relationships not exploited

We need an architecture that exploits the spatial structure of images

What We Need

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Three properties that an image-aware architecture should have:

- **Local connectivity** — each neuron connects only to a small local region of the input (the **receptive field**), not the entire image
- **Parameter sharing** — the same filter (weights) applied at every spatial location — detects the same feature everywhere
- **Translation invariance** — detecting a feature regardless of where it appears in the image

These three properties define the **convolutional layer**

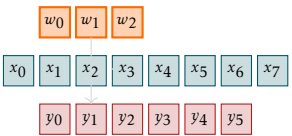
1D Convolution — Intuition

1D convolution: filter $\mathbf{w} \in \mathbb{R}^k$ slides over signal $\mathbf{x} \in \mathbb{R}^n$

$$(x * w)[i] = \sum_{j=0}^{k-1} x[i+j]\, w[j]$$

- At each position i: dot product between filter and local patch of signal
- Filter detects a specific local pattern
- Same filter applied at every position — **parameter sharing**

Example: edge detector, smoothing filter, derivative approximation

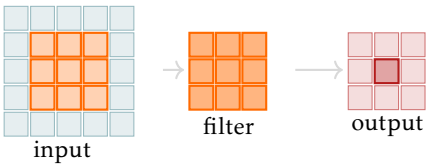


2D Convolution

2D convolution: filter $\mathbf{W} \in \mathbb{R}^{k \times k}$ slides over image $\mathbf{X} \in \mathbb{R}^{H \times W}$

$$(\mathbf{X} * \mathbf{W})[i, j] = \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} \mathbf{X}[i+p, j+q]\, \mathbf{W}[p, q]$$

- At each position (i, j): dot product between filter and $k \times k$ patch
- Output: **feature map** of shape $(H - k + 1) \times (W - k + 1)$
- Filter learns to detect a specific visual pattern (edge, corner, texture)



Stride and Padding

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Stride s: filter moves s pixels at each step

- $s = 1$: standard, output size $\approx$ input size
- $s = 2$: output halved — spatial downsampling
- Reduces computation and memory

Output size: $\lfloor (H - k)/s \rfloor + 1$

Padding p: add zeros around the border

- **Valid** ($p = 0$): no padding, output smaller than input
- **Same** ($p = (k-1)/2$): output same size as input
- Preserves spatial dimensions

Output size: $\lfloor (H + 2p - k)/s \rfloor + 1$

Multiple Channels and Filters

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RGB input: C_{in} input channels — filter must have depth C_{in}

$$(\mathbf{X} * \mathbf{W})[i, j] = \sum_{c=0}^{C_{\text{in}}-1} \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} \mathbf{X}[c, i+p, j+q]\, \mathbf{W}[c, p, q]$$

Multiple filters: C_{out} filters produce C_{out} output feature maps

- Each filter detects a different pattern
- Output tensor: $C_{\text{out}} \times H' \times W'$
- Total parameters per layer: $C_{\text{out}} \times C_{\text{in}} \times k \times k + C_{\text{out}}$ (with bias)

In PyTorch:

```
nn.Conv2d(in_channels=3, out_channels=64, kernel_size=3, stride=1, padding=1)
```

Receptive Field

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Receptive field: region of the input that influences a given neuron's output

- Layer 1 (3×3 filter): receptive field = 3×3
- Layer 2 (3×3 filter): receptive field = 5×5
- Layer L (3×3 filters): receptive field = $(2L+1) \times (2L+1)$

Key insight: deep networks have large receptive fields despite small local filters

- Early layers detect local features (edges, corners)
- Later layers combine local features into complex patterns (eyes, wheels, faces)
- Hierarchical feature learning — the hallmark of deep networks

Two 3×3 filters have the same receptive field as one 5×5 filter but fewer parameters: $2 \times 9 = 18$ vs 25

Pooling

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Pooling: spatial downsampling — reduces feature map size

Max pooling:

$$y[i,j] = \max_{p,q \in \mathcal{R}_{ij}} x[p,q]$$

- Takes maximum over local region
- Retains strongest activation
- Most commonly used

Average pooling:

$$y[i,j] = \frac{1}{|\mathcal{R}_{ij}|} \sum_{p,q \in \mathcal{R}_{ij}} x[p,q]$$

- Takes average over local region
- Smoother downsampling
- Less common in hidden layers

Benefits:

- Reduces spatial dimensions — fewer parameters in subsequent layers
- Local translation invariance — small shifts in feature position do not affect output

Global Average Pooling

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Global average pooling (GAP): averages entire feature map to a single value per channel

$$y_c = \frac{1}{H \cdot W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_c[i, j]$$

- Output: vector of length C_{out} — one value per feature map
- Replaces large FC layers at the end of the network
- Dramatically reduces parameters
- Forces each feature map to represent a single concept

Used in: ResNet, GoogLeNet, MobileNet — modern standard

In PyTorch: `nn.AdaptiveAvgPool2d((1,1))` followed by `flatten`

CNN Building Block

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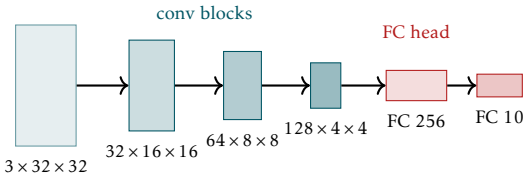
Standard pattern: Conv → BatchNorm → ReLU → (Pool)

- **Conv:** extract features via learned filters
- **BatchNorm:** stabilise activations — essential for deep CNNs
- **ReLU:** non-linearity — enables complex feature combinations
- **Pool:** spatial downsampling — reduce resolution, increase receptive field

Full CNN Architecture

Two stages: feature extraction (conv blocks) → classification (FC head)



Conv blocks:

- Increasing channels: $3 \to 32 \to 64 \to 128$
- Decreasing spatial size via pooling
- Learn hierarchical features

FC head:

- Flatten spatial features
- One or more FC layers
- Final softmax for classification

LeNet-5 (LeCun et al., 1998)

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First successful CNN — handwritten digit recognition (MNIST)

- 2 conv layers + 2 FC layers
- 28×28 input $\rightarrow$ 10 class output
- ~60k parameters — tiny by modern standards
- Used sigmoid/tanh activations, average pooling

Significance:

- Demonstrated that learned features outperform hand-crafted ones
- Introduced the conv $\rightarrow$ pool $\rightarrow$ conv $\rightarrow$ pool $\rightarrow$ FC pattern
- Deployed commercially for cheque reading

Limitation: too shallow and small for complex tasks — needed more compute

AlexNet (Krizhevsky et al., 2012)

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ImageNet moment — won ILSVRC 2012 with 15.3% top-5 error vs 26.2% runner-up

- 5 conv layers + 3 FC layers, ~60M parameters
- Key innovations:
  - ReLU activations instead of sigmoid/tanh — faster training
  - Dropout in FC layers — regularisation
  - Data augmentation — random crops, horizontal flips
  - GPU training — two GTX 580 GPUs
  - Local response normalisation (since superseded by BatchNorm)

Impact: launched the modern deep learning era — proved deep CNNs work at scale

VGG (Simonyan & Zisserman, 2014)

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Simplicity through depth — all 3×3 convolutions, increasing depth

- VGG-16: 13 conv layers + 3 FC layers, ~138M parameters
- Only 3×3 filters throughout — simplicity as a design principle
- Depth: 16–19 layers

Key insight: two 3×3 conv layers have same receptive field as one 5×5 but:

- Fewer parameters: $2 \times 3^2 = 18$ vs $5^2 = 25$
- More non-linearities — more expressive

Limitation: 3 large FC layers contain most parameters (~100M) — memory intensive

Legacy: VGG features widely used for transfer learning; clean design still instructive

ResNet (He et al., 2016)

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Problem: deeper networks were performing worse than shallower ones

Degradation problem: adding more layers hurts training accuracy — not overfitting, but optimisation failure

Key idea: **residual connections** (skip connections)

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{\mathbf{W}_i\}) + \mathbf{x}$$

- Instead of learning $\mathbf{y} = \mathcal{F}(\mathbf{x})$, learn the **residual** $\mathcal{F}(\mathbf{x}) = \mathbf{y} - \mathbf{x}$
- If optimal solution is identity, easier to learn $\mathcal{F}(\mathbf{x}) = 0$ than $\mathcal{F}(\mathbf{x}) = \mathbf{x}$
- Gradient flows directly through skip connection — no vanishing gradients

Result: successfully trained networks with 50, 101, 152 layers

Residual Block

Basic residual block:

$$\mathbf{y} = \text{ReLU}(\mathcal{F}(\mathbf{x}) + \mathbf{x})$$

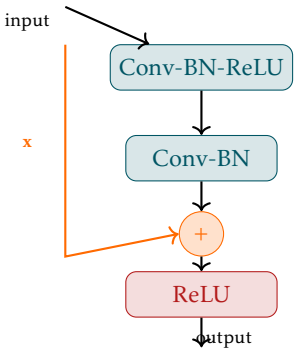
$\mathcal{F}(\mathbf{x})$: two Conv-BN-ReLU layers

Bottleneck block (ResNet-50+):

- 1×1 conv: reduce channels
- 3×3 conv: spatial processing
- 1×1 conv: restore channels

Fewer parameters than basic block at same depth

Projection shortcut: when dimensions change, use 1×1 conv on skip connection to match



1×1 Convolutions

1×1 **conv:** applies a learned linear combination across channels at each spatial location

$$y_c[i,j] = \sum_{c'=1}^{C_{\text{in}}} w_{c,c'}\, x_{c'}[i,j] + b_c$$

Equivalent to: FC layer applied independently at each spatial location

Uses:

- **Channel reduction/expansion** — change C_{in} to C_{out} cheaply (bottleneck blocks)
- **Non-linearity injection** — add ReLU between channels without spatial processing
- **Network-in-network** — increase model capacity without larger filters
- **Projection shortcuts** — match dimensions in residual connections

Depthwise Separable Convolutions

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Standard conv: $C_{\text{out}} \times C_{\text{in}} \times k \times k$ parameters

Depthwise separable conv: split into two steps

Depthwise conv:

- One filter per input channel
- Spatial processing only
- $C_{\text{in}} \times k \times k$ parameters

Pointwise conv (1×1)**:**

- Mix channels
- $C_{\text{out}} \times C_{\text{in}}$ parameters

Parameter reduction: factor of $\approx k^2$ — for 3×3: $\sim 9\times$ fewer parameters

Used in: MobileNet, EfficientNet — efficient architectures for mobile/edge deployment

Batch Norm in CNNs

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BatchNorm2d: normalises over batch **and** spatial dimensions for each channel

$$\mu_c = \frac{1}{N \cdot H \cdot W} \sum_{n,i,j} x_c^{(n)}[i,j], \qquad \sigma_c^2 = \frac{1}{N \cdot H \cdot W} \sum_{n,i,j} (x_c^{(n)}[i,j] - \mu_c)^2$$

- One mean and variance per **channel** — not per spatial location
- Learnable γ_c and β_c per channel

Placement: Conv → BN → ReLU (standard) or Conv → ReLU → BN (less common)

Effect in CNNs:

- Essential for training deep CNNs — without it, networks beyond ~10 layers are very hard to train
- Allows higher learning rates
- Reduces need for careful initialisation

Dropout in CNNs

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Standard dropout on FC layers works well in CNNs too

Problem with dropout on conv layers:

- Adjacent pixels are highly correlated
- Dropping individual activations does not create enough independence
- Information easily propagated through neighbouring non-dropped activations

SpatialDropout2d: drop entire feature maps (channels) rather than individual activations

- Removes all spatial locations of a channel simultaneously
- Forces network to use diverse feature maps
- More effective regularisation for conv layers

In practice: BatchNorm often sufficient regularisation in conv layers — dropout mainly used in FC head

Summary

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Why CNNs:

- Local connectivity
- Parameter sharing
- Translation invariance

Building blocks:

- Conv: feature extraction
- BatchNorm + ReLU: stability + non-linearity
- Pooling / stride: downsampling
- FC head: classification

Key architectures:

- LeNet → AlexNet → VGG → ResNet
- Residual connections: key to deep networks

Modern details:

- 1×1 conv: channel mixing
- Depthwise separable: efficiency
- BatchNorm2d: per-channel normalisation
- SpatialDropout: conv regularisation

In PyTorch:

- `nn.Conv2d`
- `nn.MaxPool2d`
- `nn.BatchNorm2d`
- `nn.AdaptiveAvgPool2d`

Next: L10 — Sequence Models & RNNs